QMWS Spring 2026: Introduction to Python for Data Analysis

Day Two: Extensions

Gavin Klorfine & Deborah Laze

Section 1

Importing Packages, Defining Functions

Importing Packages

- Packages add "extra stuff" to Python, including functions, constants, and more
- Some packages are installed with Python by default, and no extra work is required to use them

import math

```
import math

print(f'pi = {math.pi:.3f}')
```

```
pi = 3.142
```

Importing Packages

- We can "rename" packages to make their usage more concise

"Renaming" Packages

```
import math as m

print(f'pi = {m.pi:.3f}')
```

```
pi = 3.142
```

Importing Packages

- Some require installation. This is most easily (and safely) done through **Anaconda**
- Let's install the package **NumPy**!

Importing Packages

import numpy as np

```
import numpy as np       # Common convention

# Take a sample of 3 values from N(0, 1)
print(np.random.normal(loc = 0, scale = 1, size = 3))
```

```
[ 0.89588811 -1.35472434 -0.42418595]
```

Note

- loc $= \mu$, scale $= \sigma$

Importing Packages

- A brief aside:
  - Let's say of all the functions in **NumPy**, you only care about `numpy.random.normal`

`from ... import ...`

```
from numpy.random import normal

# Same thing, but no `np.random` at the start
normal(loc = 0, scale = 1, size = 3)
```

```
array([-0.25380965, -0.16156967,  0.31207551])
```

Defining Functions

- Sometimes you repeat the same code over and over again
  - When this happens, it is useful to collapse the repeated code into a function
  - Otherwise, code becomes long and messy due to copy-pasting

Defining Functions

- Sometimes you repeat the same code over and over again
  - When this happens, it is useful to collapse the repeated code into a function
  - Otherwise, code becomes long and messy due to copy-pasting
  - For instance, imagine needing to know how to code your own `np.random.normal()` function and then copy pasting this code every time you want to use it

Defining Functions

Defining Basic Functions

```
def addition(a, b):
    return a + b                    # What the functions spits out

print(addition(a = 1, b = 5)) # Call function

def subtraction(a, b = 1):      # Give b default value of 1
    return a - b

print(subtraction(a = 5))
```

```
6
4
```

Defining Functions

Basic Structure of a Function

```
# "arg" short for "argument"
def function_name(arg1, arg2 = default_value, arg_n):
    return output
```

Defining Functions

Function Example

```
x = [0,1,2,3,4,5,6,7]

def qm_mean(things):
    sum = 0

    for i in things:
        sum += i

    return sum / len(things)

qm_mean(x)
```

```
3.5
```

Section 2

NumPy

NumPy

- Somewhat similar to MATLAB if you have used that
- Useful for working with arrays/matrices
- Stands for **Num**erical **Py**thon
- Based in computer language **C**
  - Faster, more efficient computations

NumPy

Working With Lists/Arrays

```
import numpy as np

list1 = [1,2,3,4]
print(list1 * 2)
print(type(list1))

numpyList = np.array(list1)  # Convert list -> numpy array
print(numpyList * 2)
print(type(numpyList))
```

```
[1, 2, 3, 4, 1, 2, 3, 4]
<class 'list'>
[2 4 6 8]
<class 'numpy.ndarray'>
```

NumPy

- Matrices are typically worked with as **2D** numpy arrays

2D NumPy Arrays

```
array2d = np.array([[1,2,3], [4,5,6]]) # List of lists
print(array2d)
print(type(array2d))
```

```
[[1 2 3]
 [4 5 6]]
<class 'numpy.ndarray'>
```

NumPy

To Check Attributes of a NumPy Variable

```
array2d = np.array([[1,2,3],  # Matrices typed in a more
                    [4,5,6]]) # appealing manner

print(array2d.shape)     # Shape (# rows, # columns)
print(array2d.ndim)      # Number of dimensions
print(array2d.size)      # Number of elements
print(array2d.dtype)     # Type of elements
```

```
(2, 3)
2
6
int64
```

NumPy

- Some numpy types:
  - `uint8`
  - `int32`
  - `int64`
  - `float64`
- Generally, numpy type = type learned before + memory (in bits)

Note

- The `u` in `uint8` stands for "unsigned," meaning that the variable can only store positive values and zero.

NumPy

np.arange()

```
np.arange(start=10, stop=20, step=2, dtype = "int64")
```

```
array([10, 12, 14, 16, 18])
```

Note

- `stop` parameter is *exclusive*

np.linspace()

```
np.linspace(start=10, stop=20, num=6, dtype = "int64")
```

```
array([10, 12, 14, 16, 18, 20])
```

Tip

- Use `np.linspace()` for breaking something into equal parts
- Use `np.arange()` for creating a sequence of numbers

NumPy

- NumPy has lots of functions relating to choosing random values

Random Choice

```
print(np.random.choice(["A", "B", "C", "D"]))

print(np.random.choice(["A", "B", "C", "D"], size = 2))
```

```
C
['D' 'D']
```

NumPy

Random Numbers

```
print(np.random.normal(loc = 0, scale = 1, size = 3))
print(np.random.uniform(low = 0, high = 10, size = 3))
```

```
[-0.77502105 -0.31113943  0.91926931]
[4.82778364 0.53071542 2.02222205]
```

NumPy

- To make the values from these random functions reproducible, we set a "seed"
  - i.e., the first time the below `np.random.choice()` call is run will always return `'D'`, the second time `'B'`

Setting a Seed

```
np.random.seed(67)

# 1st time returns 'D'
print(np.random.choice(["A", "B", "C", "D"]))

# 2nd time returns 'B'
print(np.random.choice(["A", "B", "C", "D"]))
```

```
D
B
```

Indexing NumPy Arrays

```
x = np.random.normal(0, 10, size=(3,3))
print(x, "\n")  # Print x with a blank line after it

print(x[2,2], type(x[2,2]))
print(x[2,:])   # Row 2 (zero-based), all columns
```

```
[[  4.24396269   8.19276386  -4.55523498]
 [ -1.74348913 -12.91666321  -5.12083276]
 [  1.51374927   2.9382983  -14.96950941]]

-14.96950940608669 <class 'numpy.float64'>
[  1.51374927   2.9382983  -14.96950941]
```

Indexing (cont.)

```
print(x, "\n")

print(x[2, 0:2])    # Row 2, columns 0 and 1
print(type(x[2, 0:2]))
```

```
[[  4.24396269   8.19276386  -4.55523498]
 [ -1.74348913 -12.91666321  -5.12083276]
 [  1.51374927   2.9382983  -14.96950941]]

[1.51374927 2.9382983 ]
<class 'numpy.ndarray'>
```

NumPy

- You may also perform **aggregate** functions on numpy arrays

NumPy Aggregate Functions

```
x = np.array([1,4,7,15])

print(np.mean(x))
print(np.median(x))
print(np.std(x))
```

```
6.75
5.5
5.2414634036535525
```

More Aggregate Functions

```
x = np.array([1,4,7,15])

print(np.min(x))          # Minimum value
print(np.argmin(x))       # Position of minimum value
print(np.max(x))          # Maximum value
print(np.argmax(x))       # Position of maximum value
```

```
1
0
15
3
```

NumPy

Check-in

- How might you generate 100 random values from the standard normal distribution?
- How might you get the maximum of these values?

NumPy

Check-in

- How might you generate 100 random values from the standard normal distribution?
- How might you get the maximum of these values?

```
np.max(np.random.normal(loc=0, scale=1, size=100))
```

```
np.float64(1.6230497183839991)
```

Section 3

pandas

pandas

- For the analysis of spreadsheet-like data in Python
- Somewhat similar to **R** if you have used that
  - Introduces the `DataFrame`

pandas

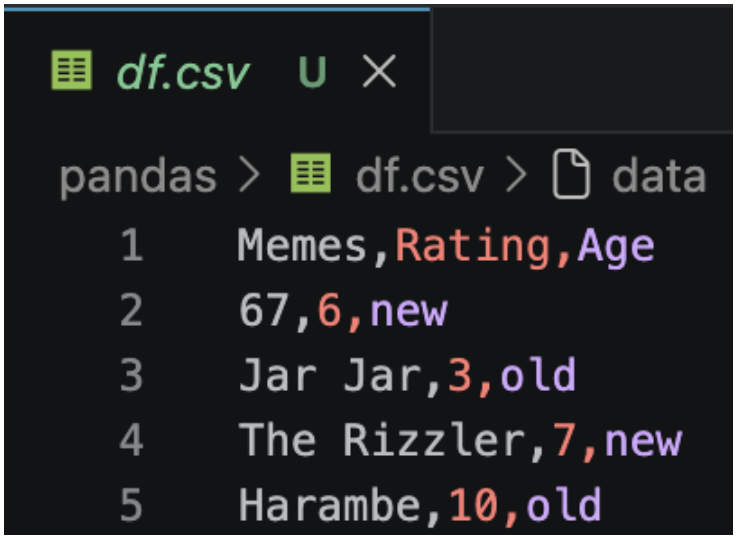
Creating a DataFrame

```
import pandas as pd
df = pd.DataFrame({
    "Memes" : ["67", "Jar Jar", "The Rizzler", "Harambe"],
    "Rating": [6, 3, 7, 10],
    "Age"   : ["new", "old", "new", "old"]
}, index = ['a','b','c','d']) # Index is optional
df # To print
```

	Memes	Rating	Age
a	67	6	new
b	Jar Jar	3	old
c	The Rizzler	7	new
d	Harambe	10	old

pandas

- Data are commonly stored as `.csv` files
  - **C**omma **s**eparated **v**alues
- These files have entries separated by commas, often including a first row of "header" entries
- A `.csv` file of the **pandas** `DataFrame` from the previous slide (variable `df`) is shown below



pandas

- Continuing with the `df DataFrame`, the below shows how a `.csv` file is written

Writing `.csv` Files

```
# Write df.csv to the pandas/ folder
df.to_csv("pandas/df.csv", index = False)
```

- Here, `to_csv` is a **method** acting on the `df` **object** (or variable)

Tip

Setting the `index` argument to `False` removes row indices (i.e., row 0, 1, 2, ..., n) from being stored as a separate column

pandas

- We often need to do the reverse; namely, reading data from a `.csv` file into a **pandas** `DataFrame`

Reading `.csv` Files

```
# Read df.csv from the pandas/ folder
df_new = pd.read_csv("pandas/df.csv")

df_new
```

	Memes	Rating	Age
0	67	6	new
1	Jar Jar	3	old
2	The Rizzler	7	new
3	Harambe	10	old

pandas

- **pandas** also has functions to read/write data from other file formats, such as from Microsoft Excel `.xlsx` files

`.xlsx` Files

```
excelDat = pd.read_excel( # Read
    "file_name.xlsx", sheet_name = "optional"
)

excelDat.to_excel( # Write
    "folder/excelDat.xlsx", sheet_name = "optional"
)
```

Inspecting Data

```
df.head(2) # First 2 rows
```

	Memes	Rating	Age
a	67	6	new
b	Jar Jar	3	old

```
df.tail(2) # Last 2 rows
```

	Memes	Rating	Age
c	The Rizzler	7	new
d	Harambe	10	old

Inspecting Data

```
df.info()
```

```
<class 'pandas.DataFrame'>
Index: 4 entries, a to d
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   Memes   4 non-null      str
 1   Rating  4 non-null      int64
 2   Age     4 non-null      str
dtypes: int64(1), str(2)
memory usage: 128.0+ bytes
```

Descriptives

```
df.describe()
```

	Rating
count	4.000000
mean	6.500000
std	2.886751
min	3.000000
25%	5.250000
50%	6.500000
75%	7.750000
max	10.000000

Grouping

```
df.groupby("Age").describe()
```

	Rating							
	count	mean	std	min	25%	50%	75%	max
Age								
new	2.0	6.5	0.707107	6.0	6.25	6.5	6.75	7.0
old	2.0	6.5	4.949747	3.0	4.75	6.5	8.25	10.0

Selecting a Column

```
df["Memes"]
```

```
a              67
b         Jar Jar
c     The Rizzler
d         Harambe
Name: Memes, dtype: str
```

A Note on Types

```
type(df["Memes"])
```

`pandas.Series`

- A variable of type `pandas.Series` is essentially a column of a **pandas** `DataFrame`

- To select multiple columns, provide a list of column names as the index

Selecting Multiple Columns

```
df[["Memes", "Age"]]
```

	Memes	Age
a	67	new
b	Jar Jar	old
c	The Rizzler	new
d	Harambe	old

- A simple way to filter rows is by row number:

Filter Rows by Row Number

```
df[0:2] # Row numbers 0 and 1
```

	Memes	Rating	Age
a	67	6	new
b	Jar Jar	3	old

- You can also filter rows by indexing a `DataFrame` with a boolean expression involving the same `DataFrame`

Filter Rows With Boolean Expressions

```
df[df["Age"] == "old"]
```

	Memes	Rating	Age
b	Jar Jar	3	old
d	Harambe	10	old

- The result of the previous slide was also a `DataFrame`!
  - This means we can index it further by, say, taking the `"Memes"` column:

Indexing Twice

```
df[df["Age"] == "old"]["Memes"]
```

```
b    Jar Jar
d    Harambe
Name: Memes, dtype: str
```

pandas

- This also means that methods shown previously can be applied to these filtered `DataFrames`

Describing Filtered Data

```
df[df["Age"] == "old"].describe()
```

	Rating
count	2.000000
mean	6.500000
std	4.949747
min	3.000000
25%	4.750000
50%	6.500000
75%	8.250000
max	10.000000

pandas

- We can also select rows and columns using the `loc` or `iloc` **attributes**
  - These are similar to methods but use square `[]` brackets (instead of round brackets `()`)

Selecting with `loc` and `iloc`

```
df.loc["b", "Memes"] # Row b of Memes column
```

```
'Jar Jar'
```

```
df.iloc[1,0] # Row 1 of 0th column (Memes)
```

```
'Jar Jar'
```

Note

A brief note on terminology: - The distinction between *methods* and *attributes* is that an attribute gives access to information associated with a variable; whereas a method is a function that may act on/with the variable. + E.g., `.loc` versus `.mean()`

- Like **NumPy**, we can use a colon (`:`) with `loc` and `iloc` to indicate "all rows" or "all columns":

Selecting All Rows with `loc`

```
df.loc[:, "Memes"] # All rows of Memes column
```

```
a              67
b         Jar Jar
c     The Rizzler
d         Harambe
Name: Memes, dtype: str
```

Selecting All Columns with `iloc`

```
df.iloc[0] # All columns for 0th row (row 'a')
```

```
Memes      67
Rating      6
Age       new
Name: a, dtype: object
```

Note

- If only the row argument is supplied, all columns are selected by default
  - This is also true for `loc`

- If you would like to select multiple columns (or rows), but not all columns (or rows), pass a `list` to `loc` or `iloc`

Selecting Multiple (Specific) Columns with `loc`

```
df.loc[["a", "b"], ["Memes", "Age"]]
```

	Memes	Age
a	67	new
b	Jar Jar	old

- Same thing for `iloc`, but with row numbers instead of indices/names

- We can also use boolean expressions with `loc` (but not `iloc`):

Boolean Expressions with `loc`

```
# Take rows where age is old and just the meme column
df.loc[df["Age"] == "old", "Memes"] # Old memes!
```

```
b    Jar Jar
d    Harambe
Name: Memes, dtype: str
```

- In these boolean expressions, we can use `&` for `and`, `|` for `or`, and `~` for `not`
  - We need these special operators as we are comparing entire columns as opposed to single values
  - We also surround expressions to be compared in round brackets `()`

Boolean Expressions with `loc`

```
# Take all rows where age is old and rating is larger than 7
df.loc[(df["Age"] == "old") & (df["Rating"] > 7), :]
```

	Memes	Rating	Age
d	Harambe	10	old

pandas

	Memes	Rating	Age
a	67	6	new
b	Jar Jar	3	old
c	The Rizzler	7	new
d	Harambe	10	old

How might I index rows with rating greater than or equal to seven?

- I want 2 different answers! (Hint: `loc` and `df[?]`)

How might I index rows with rating greater than or equal to seven?

```
df[df["Rating"] >= 7]
df.loc[df["Rating"] >= 7, :] # The `, :` is optional
```

	Memes	Rating	Age
c	The Rizzler	7	new
d	Harambe	10	old

pandas

- To show a peculiarity of pandas, I first set the index of our data frame `df` (e.g., the row names `"a"`, `"b"`, ...) back to integers (the default)

Changing the Index

```
df.index = [0,1,2,3]
df
```

	Memes	Rating	Age
0	67	6	new
1	Jar Jar	3	old
2	The Rizzler	7	new
3	Harambe	10	old

- When using `loc` on a data frame with a numeric index, the format of a slice is `inclusive:inclusive`

`loc` is Strange

```
df.loc[0:1] # inclusive:inclusive
```

	Memes	Rating	Age
0	67	6	new
1	Jar Jar	3	old

- For reference, here is the same line but using `iloc` instead of `loc`:

`iloc` is Typical

```
df.iloc[0:1] # inclusive:exclusive
```

	Memes	Rating	Age
0	67	6	new

- We can also add columns to our data frame. Here I add a column `r10`: the `Rating` out of 10

Adding a Column

```
df["r10"] = df["Rating"] / 10
df
```

	Memes	Rating	Age	r10
0	67	6	new	0.6
1	Jar Jar	3	old	0.3
2	The Rizzler	7	new	0.7
3	Harambe	10	old	1.0

pandas

- We might also want to sort these entries by `Rating`

Sorting a Data Frame

```
df.sort_values(by = "Rating", inplace = True)
df
```

	Memes	Rating	Age	r10
1	Jar Jar	3	old	0.3
0	67	6	new	0.6
2	The Rizzler	7	new	0.7
3	Harambe	10	old	1.0

- Argument `inplace = True` modifies `df` directly instead of returning a new one

- We can also sort by `Age` first and then `Rating`
  - Strings are sorted alphabetically

Sorting by a List

```
df.sort_values(by = ["Age", "Rating"], inplace = True)
df
```

	Memes	Rating	Age	r10
0	67	6	new	0.6
2	The Rizzler	7	new	0.7
1	Jar Jar	3	old	0.3
3	Harambe	10	old	1.0

pandas

- Instead of using the `describe()` method, you may want only a select few (or one) statistic

Getting Specific Descriptive Statistics

```
df["Rating"].mean()
```

```
np.float64(6.5)
```

```
df["Rating"].std()
```

```
np.float64(2.886751345948129)
```

Getting Specific Descriptive Statistics

```
df["Rating"].agg(['mean', 'std']) # agg = aggregate
```

```
mean    6.500000
std     2.886751
Name: Rating, dtype: float64
```